

SolaX Inverter Solution: X3 Hybrid G4 + External Wind Turbine Solution



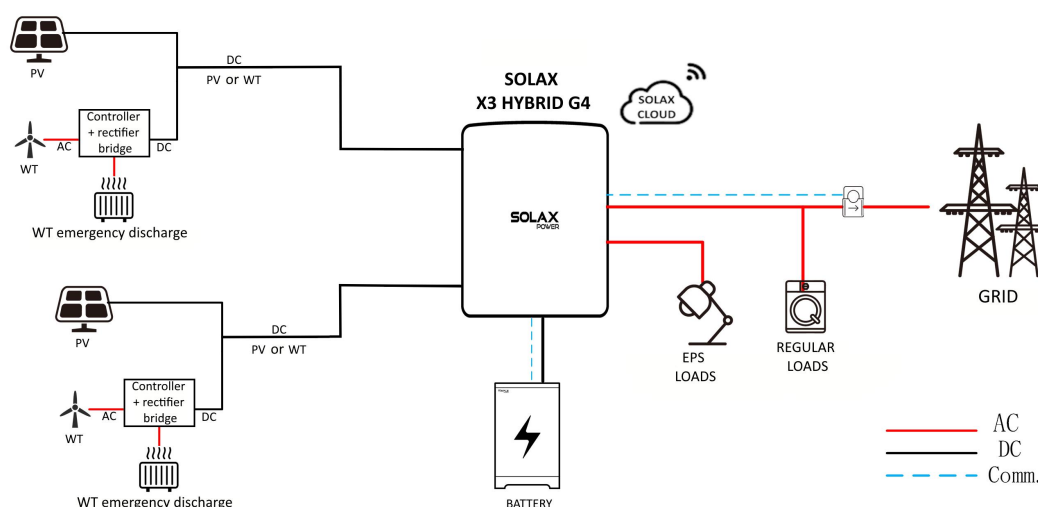
Introduction

With the growing global demand for renewable energy, alongside PV (photovoltaic power), wind energy is becoming a key element of modern energy systems. The integration of energy storage with wind turbines (WT) helps reduce the instability and variability of wind power generation, improving self-consumption and the overall reliability of the system.

Many regions of Europe, including Poland, support the development of wind energy through policies such as subsidies for wind turbines, aimed at increasing the share of RES and accelerating the energy transition.

In this context, selected SolaX inverters have been designed to convert wind energy into useful electricity for household consumers, integrating functions that allow operation with a wind turbine.

This document presents the three-phase hybrid inverter from the X3 HYBRID G4 series, illustrated with the X3 Hybrid G4 15K as a practical example.



the X3 Hybrid G4 wind turbine solution

In the X3 Hybrid G4 inverter, both PV1 and PV2 inputs (which are MPPT inputs) can be connected to wind turbines (WT), but only through a controller equipped with protective circuits.

This solution allows flexible use of the PV1 and PV2 inputs to connect one or even two wind turbines, or one turbine and second one to PV panels

Warranty terms for the inverter when used with a wind turbine

To maintain the inverter warranty, the following rules must be strictly observed when integrating wind turbines or other energy sources:

1. **Use Only Certified Controllers and Components:**
Connect wind turbines only using Wind Turbine manufacturer-designated controllers equipped with appropriate automatic protection functions. Using incompatible devices may result in warranty loss.
2. **Respect Electrical Limits:**
Do not exceed the inverter's maximum input voltage, current, and power ratings for PV inputs used with wind turbines. Overloading or improper configuration can cause damage not covered by warranty.
3. **Proper Installation and Grounding:**
Ensure all system components—including turbines, controllers, inverters, and resistive dump loads—are installed according to manufacturer guidelines and correctly grounded to avoid faults.
4. **Use Appropriate Rectification:**
Always convert the wind turbine's AC output to DC before feeding the inverter's PV inputs, using a suitable rectifier bridge as specified.
5. **Follow Configuration Guidelines:**
Set inverter input parameters (such as power curve VA points, voltage, and current limits) precisely as recommended in the product documentation.
6. **Avoid Unauthorized Modifications:**
Do not use any software other than the official Solax software, do not physically modify the inverter or its equipment, and use only accessories (e.g., plugs) provided with the inverter.
7. **Perform Regular Maintenance and Monitoring:**
Keep the system properly maintained and monitored for faults or abnormal operation to comply with warranty terms.
8. **Comply with the following Technical Requirements as well as the general warranty conditions of Solax devices.**

Technical Requirements

1. Wind Turbine Controller / Self-Breaker Requirement

Wind turbines must have a dedicated controller and/or self-breaker.

Once:

a) the load power demand drops, the batteries are full, and grid export is not allowed,

or

b) the inverter must shut down due to grid parameter faults or other reasons,

Then:

the inverter will stop using wind turbine (WT) power. As a result, the WT speed and

voltage may increase rapidly, which can be extremely dangerous.

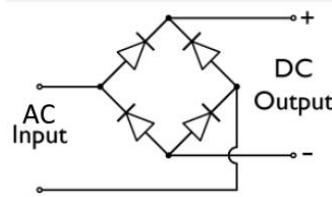
Therefore, the wind turbine (WT) controller must have at least automatic protection capabilities to regulate the voltage and power produced by the WT by unloading excess energy to an emergency self-breaker, usually a resistive load. Such control functions ensure the safe connection of the wind turbine to the inverter.



2. DC Input Requirement on Inverter PV Ports

The inverter's PV inputs accept only DC voltage and current.

Thus, the AC voltage generated by the wind turbine must first be converted to DC via an appropriate rectifier bridge before connecting to the inverter.



3. Electrical limitations for connecting a wind turbine to X3 Hybrid G4 15K

When connecting a wind turbine to PV1 or PV2 input of the X3 Hybrid G4 15K inverter, the following limitations must be observed:

- Maximum DC voltage from WT: 600 V
- Maximum current from WT on PV1: 20 A
- Maximum current from WT on PV2: 10 A
- Maximum power: 10 kW on PV1, 5 kW on PV2
- Minimum starting voltage: 150 V
- Minimum stop voltage: 120 V

VA 12	
Voltage(V)	Current(A)
750.0	250.00
Value range 0.0~600.0	Value range 0.00~20.00
Save	

4. Configuration of the PV input for operation with a wind turbine

Both PV inputs (PV1 and PV2) can be activated for operation with a wind turbine. Each has separately programmable VA points (voltage-current), which enables:

- a) Connecting a wind turbine to one of the PV inputs while using the other for a traditional PV string.
- b) Connecting two different wind turbines—each to a separate PV input.

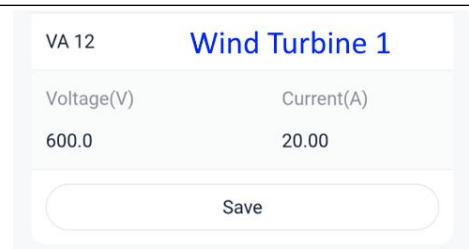
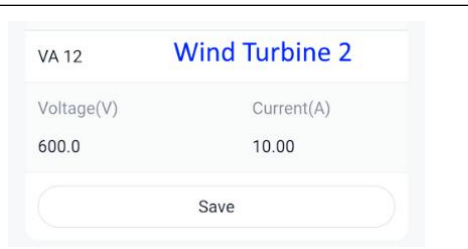
It should be noted that the VA points vary among different types, powers, and manufacturers of wind turbines and must be specified by the turbine manufacturer.

The table below is not real VA curve list, shows only the maximum reference current and power values at various voltages for the PV input used as the wind turbine input for PV1 and PV2 in X3-Hybrid-G4 15K.

	Voltage (V)	Max. Current (A)		Max Power (W)	
	PV1&PV2	PV1	PV2	PV1(2 string)	PV2
VA1	120	20	10	2400	1200
VA2	150	20	10	3000	1500
VA3	170	20	10	3400	1700
VA4	200	20	10	4000	2000
VA5	250	20	10	5000	2500
VA6	300	20	10	6000	3000
VA7	350	20	10	7000	3500
VA8	400	20	10	8000	4000
VA9	450	20	10	9000	4500
VA10	500	20	10	10000	5000
VA11	550	20	10	10000 (blocking)	5000 (blocking)
VA12	600	20	10	10000 (blocking)	5000 (blocking)

5. Power Curve Last Point Setting (VA12)

The last point of the power curve (VA12) should be set to the maximum allowed input values:

600 V and 20 A for PV1 / Wind Turbine 1	600 V and 10 A for PV2 / Wind Turbine 2
	

This setting is used as a blocking or breaking point for the wind turbine to allow the

inverter to produce energy for as long as possible + to protect the system.

6: Wind Turbine Controller Brake Resistor Activation Voltage

The wind turbine controller should start sending power to the brake resistor at a voltage max 10 V higher than the inverter's maximum voltage.

If the controller starts working below the inverter's maximum voltage, set the inverter's maximum voltage and current limits 10 V lower than the controller's start voltage.



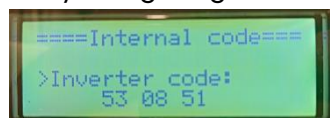
Setting Wind Turbine function

1. Pre-Connection Setup Instructions

Before physically connecting the wind turbine to the inverter via the PV1 or PV2 terminal, you must:

- a) Ensure that the inverter is updated to software version DSP_V1.53 and ARM_V1.51 (internal code: 53 xx 51 or higher).

You can find the internal code by navigating to: Menu → About → Internal Code



- b) For remote settings, the Solax communication module Pocket 3.0 series (or higher) must also be updated to the latest available firmware version.

Note: Only inverters compatible with the black Pocket 3.0 can implement remote settings for operation with wind turbines (WT).



- c) complete the following setup — either manually through the LCD screen or remotely.

Note:

To perform the update and configure the wind turbine (WT) settings in the X1 MINI G4, the inverter must have:

- AC voltage from Grid
- Stable at least 180 V DC on the MPPT or battery port

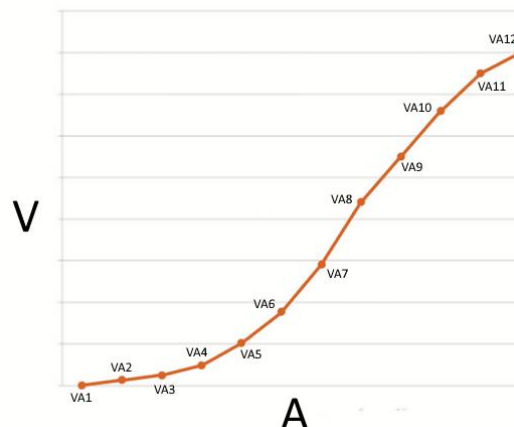
Therefore, during the update and configuration, it is required to provide a stable DC voltage source (e.g., from a DC power supply, PV simulator with appropriate parameters).



Solar	
PV1 Voltage	180.9V
PV2 Voltage	0.0V
PV1 Current	0.2A
PV2 Current	0.0A
PV1 Input Power	37.0W
PV2 Input Power	0.0W
Grid	
Voltage 1	230.4V
Voltage 2	232.6V
Voltage 3	232.6V

2. Guidelines for setting the VA curve

- The VA (Volt-Amper) curve defines the allowable relationships between voltage and current for individual operating points of the wind turbine connected to the inverter.
- Proper configuration of the VA curve helps protect the system and optimize energy production.
- The optimal VA curve is specified by the wind turbine manufacturer but should always be entered considering the permissible parameters of the inverter (see section 3 Technical Requirements).



2.1 Key principles for the VA curve of the discussed X3 HYBRID G4 15K:

- The voltage points must increase progressively: $V1 < V2 < V3 \dots < V12$.
- The voltage range for each point should be within: 120 V – 600 V.
- Current values (A1 ~ A12) can be set independently but should not exceed:
PV1: 20 A

PV2: 10 A

- d) The last point of the curve (VA12) should be set to the maximum allowable input values according to the inverter's limitations (voltage, current, and rated power, see section 3 Technical Requirements).

It is recommended to set the initial points with low current to avoid excessively inhibiting the turbine at startup.

VA Point	Voltage [V]	Current PV1 [A]	Current PV2 [A]
VA1	120	0,5	0,5
VA2	150	1	1
...	...	...	...
VA12	600	20	10

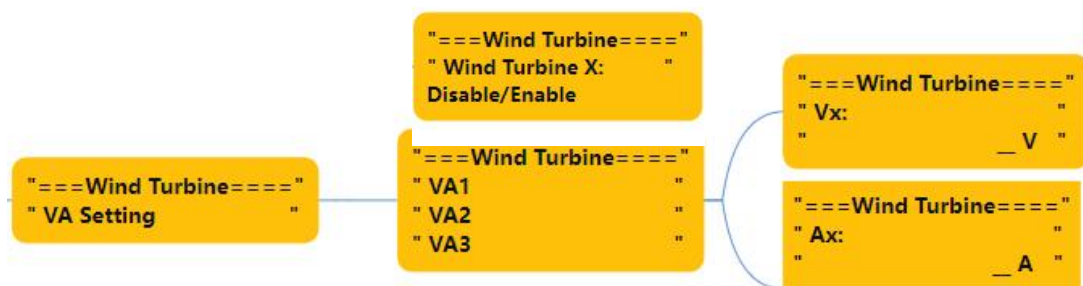
Sample VA points table for X3 Hybrid G4

2.2 Additional Recommendations:

- Follow the wind turbine manufacturer's instructions regarding declared VA points.
- Configured values should match the actual parameters of the specific wind turbine model used.
- After saving the settings, it is recommended to verify them by monitoring the inverter operation.

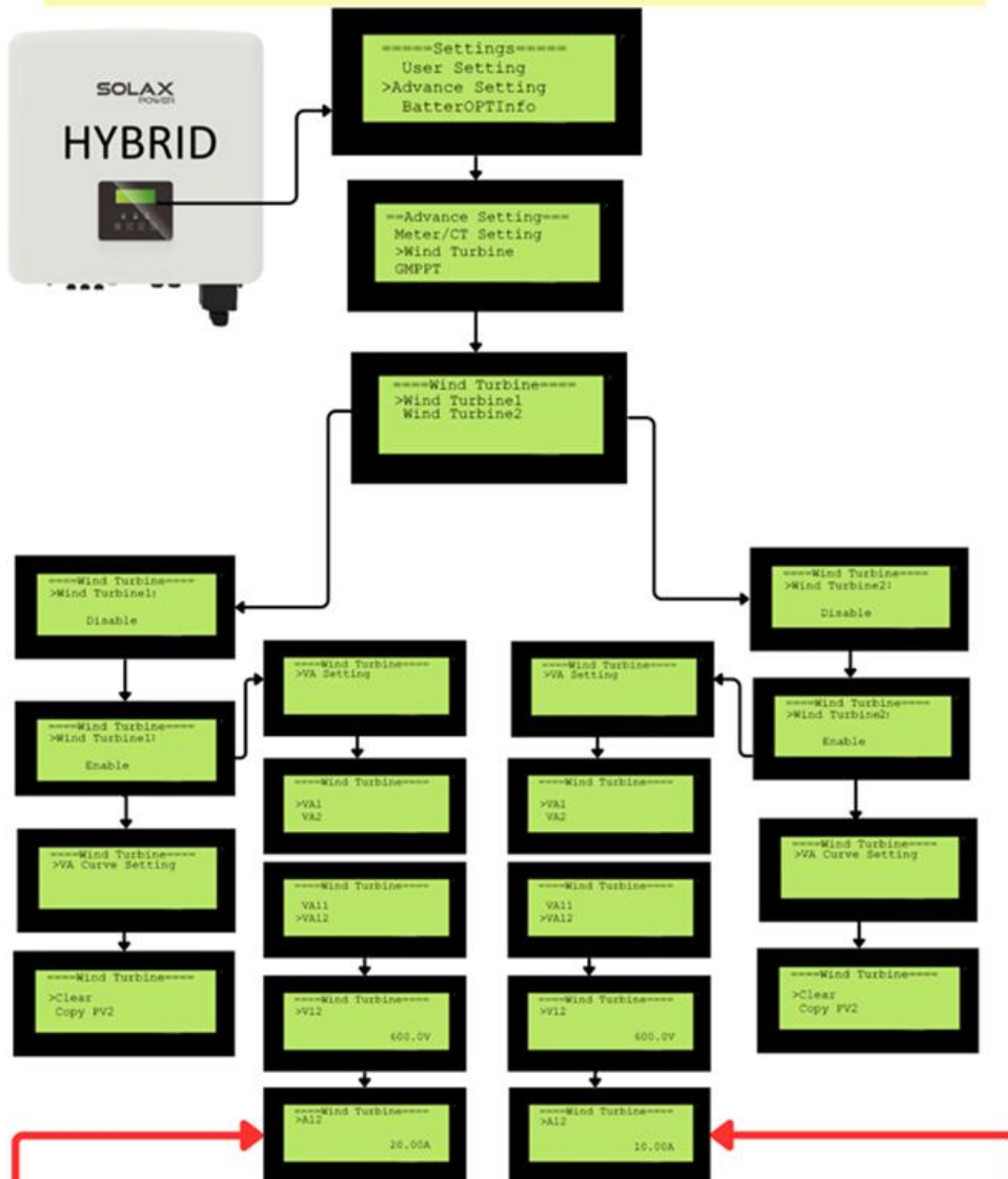
2.3 Manual settings

- Select Settings (installer code: 2014) > Advanced Settings > Wind Turbine1/ Wind Turbine2, and set Wind Turbine1 and/or Wind Turbine2 to: Enabled.
- For the PV input activated for the Wind Turbine, select the VA Setting, where the



VA points (VA1-VA12) are set separately for each input

Manual settings on inverter LCD menu



Set the last point of the power curve (VA12) to the maximum allowed input values:

- 600 V i 20 A for PV1/ Wind Turbine1
- 600 V i 10 A for PV2/ Wind Turbine2

This helps to safely break WT at highest parameters but still producing power.

Remote settings in advanced settings:

The Solax Cloud (V6.9.1 or higher) app also supports remote setup of WT.

For these settings, the inverter must be online, connected to the end-user account, and updated together with the communication module.

Remote settings are more convenient and recommended to use.

The image shows two screenshots from the Solax Cloud app. The left screenshot shows the 'Advanced' settings menu with 'Wind Turbine' selected. The right screenshot shows the 'Wind Turbine' settings for 'Wind Turbine1'. The 'Function Control' is set to 'Enable'. Below this, there are 12 points (VA 1 to VA 12) for the power curve. The last point, VA 12, is set to 600.0 V and 20.00 A. A yellow arrow points from the 'Wind Turbine' option in the left menu to the 'Wind Turbine' settings screen on the right. Another yellow arrow points from the 'Confirm' button to the 'VA 12' point.

Remember: set the last point of the power curve (VA12) to the maximum allowed input values:

- 600 V and 20 A for PV1/ Wind Turbine 1,
- 600 V and 10 A for PV2/ Wind Turbine 2.

This helps to safely break WT at highest parameters but still producing power.

Final remarks

Declared allowed parameters of X3 Hybrid G4 for WT work						
Model	Nominal output power	PV1 Max power	PV2 Max power	Max voltage per PV input	PV1 Max current	PV2 Max current
X3-Hybrid-5.0-D	5K	5K	5K	600V	10A	10A
X3-Hybrid-6.0-D	6K	5K	5K		10A	10A
X3-Hybrid-8.0-D	8K	10K	5K		20A	10A
X3-Hybrid-10.0-D	10K	10K	5K		20A	10A
X3-Hybrid-12.0-D	12K	10K	5K		20A	10A
X3-Hybrid-15.0-D	15K	10K	5K		20A	10A
Caution: wind turbine nominal power must be smaller or equal to inverter nominal output + must not exceed the permissible power per input				Always set the last maximum VA point so that the obtained power does not exceed the inverter's rated power.		

Supported inverter models and software version for local and remote settings:

Model	Progress	Software version
X3-Hybrid-G4	ready	DSP_V1.53; ARM_V1.51
X1-MINI-G4	ready	DSP_V028; ARM_V035
Solax Cloud	ready	V 6.9.1
Communication modules Pocket 3.0 (black housing)	ready	Upload latest software

Safety Operation Notes

1. Before starting the installation, ensure that the wind turbine (WT) is powered off before making any connections.
2. Verify that the grounding of the wind turbine, controller, and inverter is fully secure and reliable.

Note:

information provided in this manual (V1.6EN dated 22.08.2025) may change in the future.

Please always use the latest available version of the manual.